PAMCA503 · DATABASE MANAGEMENT SYSTEMS · COURSE PROJECT

Restaurant Management System

Ember — a database-driven web application

STUDENT

Aryan Ayush · 26MCA0009

FACULTY

Dr. M. Rajkumar

PROGRAMME

M.C.A. · Course Project

Aryan Ayush · 26MCA0009

From paper registers to one database

THE PROBLEM

01 Double-booked tables

02 Lost or misread orders

03 Billing mistakes

04 No member history or sales reports

ABSTRACT

One MySQL database holds the whole restaurant. A customer website and a staff panel read and write the same tables, so every screen always agrees.

11 related tables in one database: restaurantdb

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Two front-ends, one database

CUSTOMER WEBSITE

Guests and members

- Menu, cart and PDF bill
- Table reservations
- Member register and login
- Profile with loyalty points



STAFF PANEL

Staff after login

- Point of sale and kitchen
- Bills and tables
- Menu, staff and accounts
- Sales and revenue reports

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Three tiers, one request



1 Browser asks for menu.php

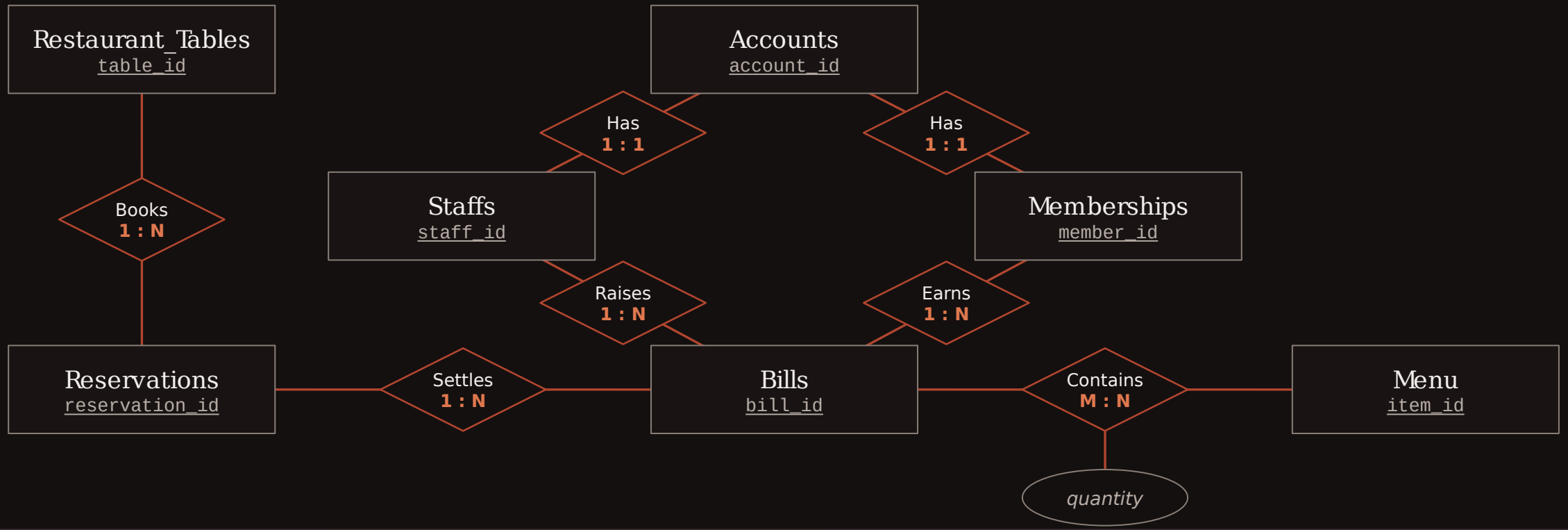
2 PHP queries MySQL `SELECT * FROM Menu`

3 MySQL returns the rows

4 PHP builds HTML, browser shows the page

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ER diagram & schema



11 **TABLES IN THE SCHEMA** Menu · Accounts · Staffs · Memberships · Restaurant_Tables · Reservations · Bills · Bill_Items · Table_Availability · Kitchen · card_payments

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Keys, constraints & normalization

FROM restaurantDB.sql

```
CREATE TABLE Memberships (
  member_id   INT PRIMARY KEY AUTO_INCREMENT,
  member_name VARCHAR(255),
  points      INT,
  account_id  INT,
  FOREIGN KEY (account_id)
    REFERENCES Accounts(account_id)
);
```

CONSTRAINTS

PRIMARY KEY · FOREIGN KEY · AUTO_INCREMENT · NOT NULL

DATA TYPES

VARCHAR · INT · DECIMAL · DATE · TIME · DATETIME

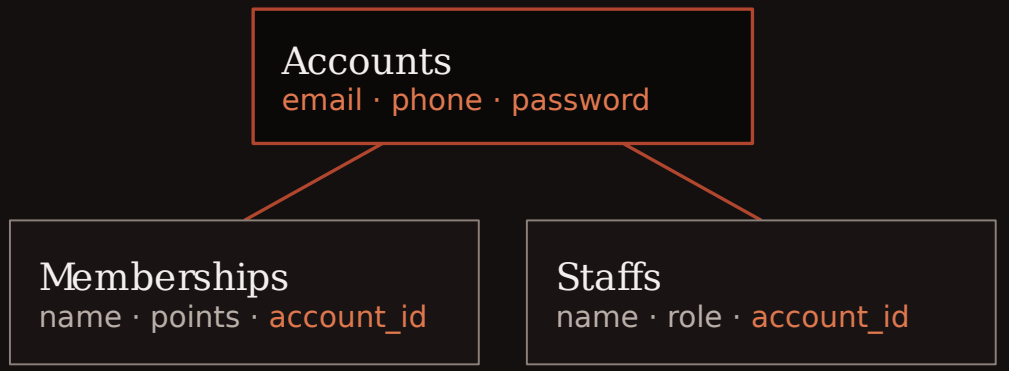
WITHOUT NORMALIZATION

Memberships
name · points · email · phone · password

Staffs
name · role · email · phone · password

login details repeated in two tables

IN EMBER · STORED ONCE



Same idea: Bills + Bill_Items — one row per dish, no dish1, dish2 … columns.

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Real queries from the project

FREE TABLES

Big enough and not already booked

reservePage.php

```
SELECT * FROM Restaurant_Tables
WHERE capacity >= '$head_count'
  AND table_id NOT IN ($reserved_table_ids_string)
```

JOIN + AGGREGATE

Today's revenue from three tables

statistics-panel.php

```
SELECT SUM(item_price * quantity) AS total_revenue
FROM Bill_Items
INNER JOIN Menu  ON Bill_Items.item_id = Menu.item_id
INNER JOIN Bills ON Bill_Items.bill_id = Bills.bill_id
WHERE DATE(Bills.bill_time) = '$currentDate'
```

TRANSACTION

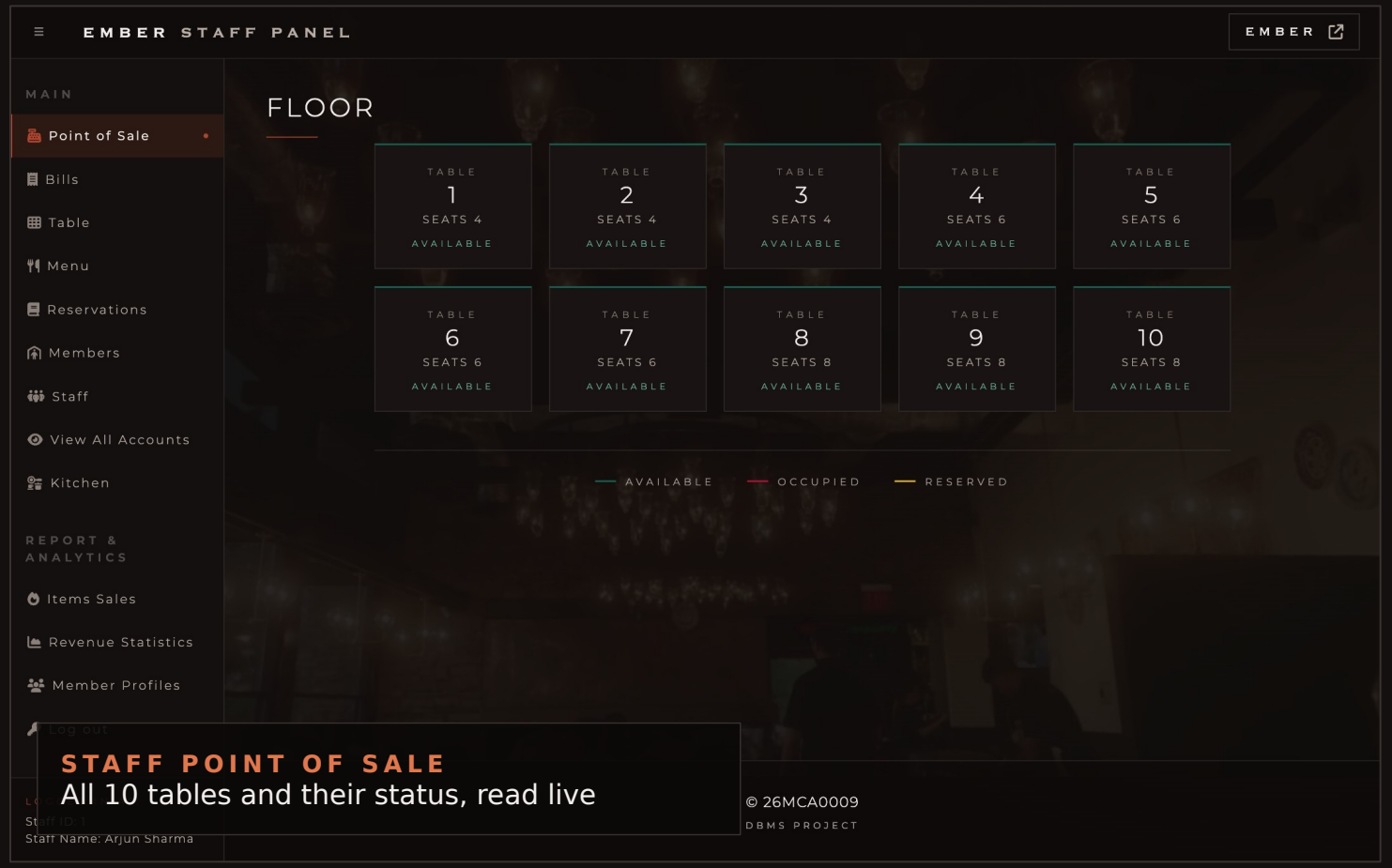
Register = INSERT Accounts + INSERT Memberships → COMMIT, or ROLLBACK if one fails

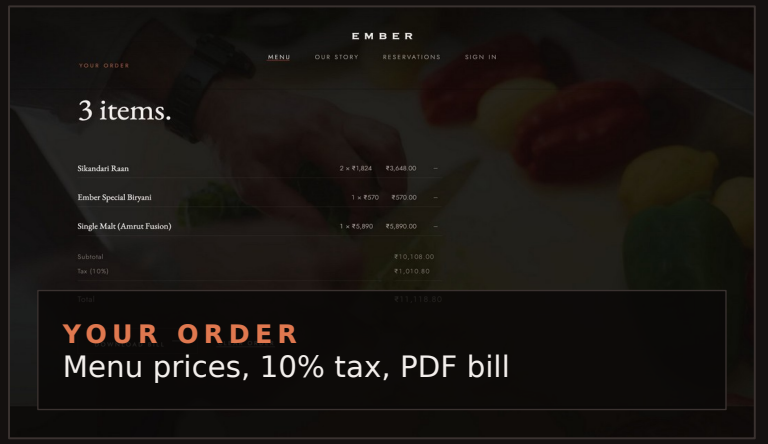
PREPARED STATEMENTS

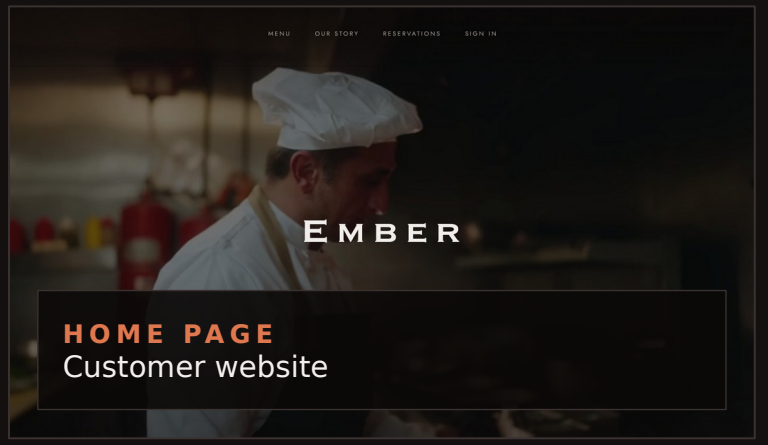
? placeholders on login, register, cart and menu lookups stop SQL injection

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The running application







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What the database makes possible

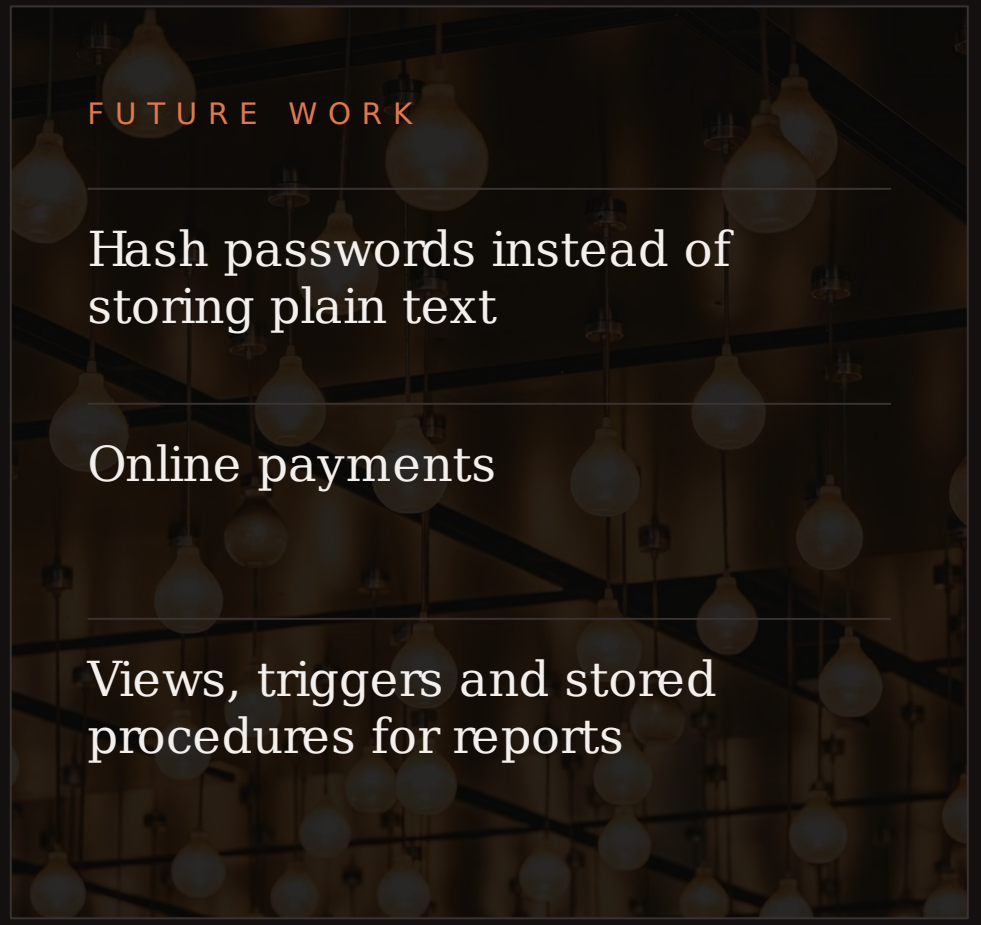
TAKEAWAYS

01 One consistent database

Bookings, orders, bills, members and staff stay in sync.

02 Real DBMS concepts in a live app

Keys, joins, aggregates and transactions.

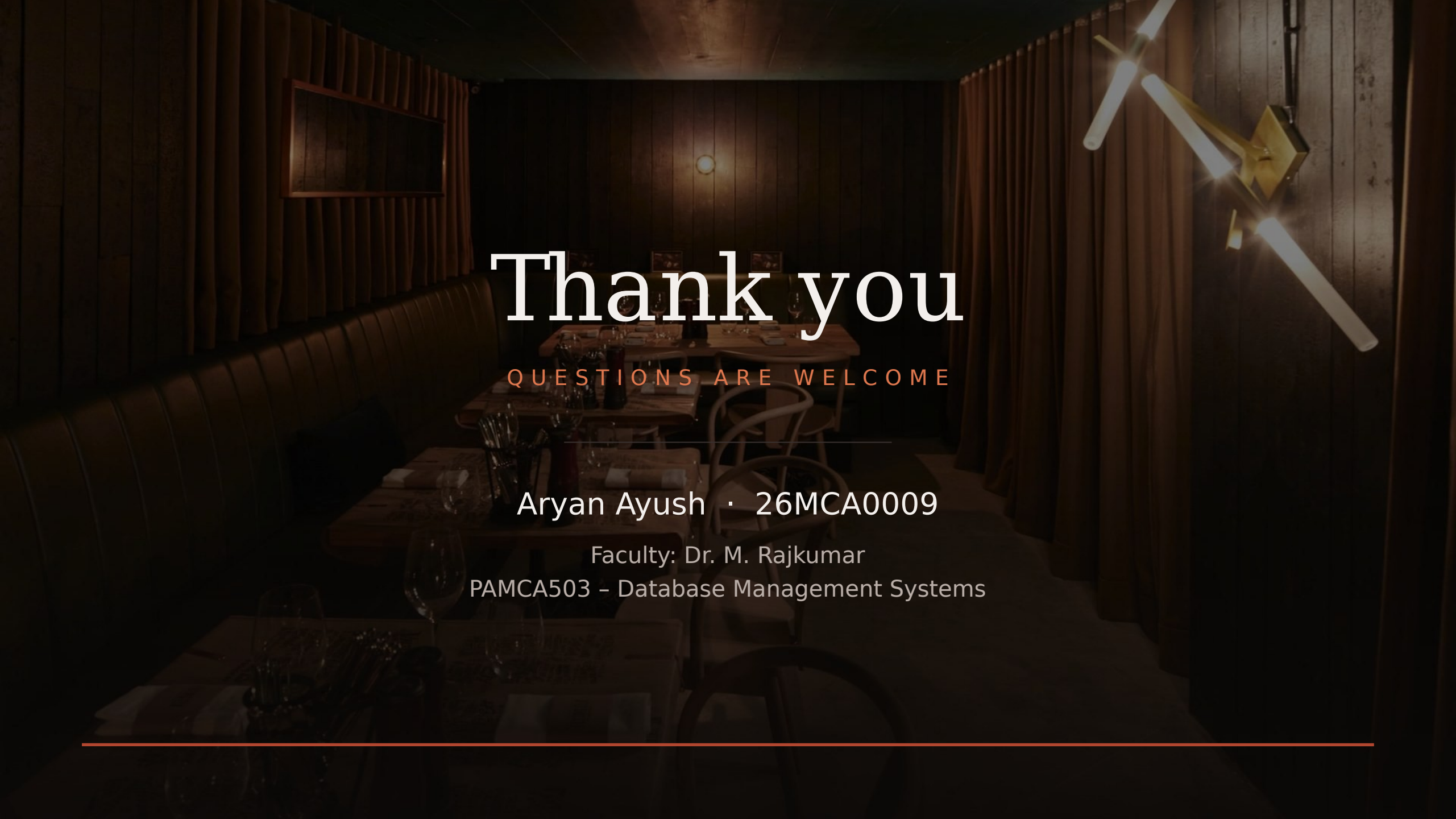


FUTURE WORK

Hash passwords instead of storing plain text

Online payments

Views, triggers and stored procedures for reports



Thank you

QUESTIONS ARE WELCOME

Aryan Ayush · 26MCA0009

Faculty: Dr. M. Rajkumar

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